

Project

- QuizMaster (Web and Mobile Application)

Related Files

- [QuizMaster API Endpoints](#)
- [QuizMaster UI/UX Design\(Figma\)](#)
- [QuizMaster Diagram \(Diagrams.NET\)](#)
- [QuizMaster Microservices Architecture](#)
- [QuizMaster Web App Frontend](#)
- [HowToPost Question](#)
- [HowToStart Session \[Frontend and Backend connectivity\]](#)
- [Progress Tracker](#)

Vision

"To ignite the spirit of friendly competition, knowledge exploration, and personal growth through the Quiz Bee Competition Web and Mobile Application, creating a community of lifelong learners and champions."

Mission

"At QuizMasters, our mission is to provide a dynamic and engaging platform for individuals of all ages and backgrounds to come together, test their knowledge, and celebrate learning. We strive to make the QuizMasters Web and Mobile Application a space where participants can challenge themselves, showcase their expertise, and learn from others. Through this platform, we aim to foster a sense of camaraderie, encourage intellectual development, and inspire a thirst for knowledge in an inclusive and accessible manner."

Success Criteria

- **Community Flourishing:** Measure the growth of our Quiz Bee community, with a focus on diverse participation from individuals from different parts of the country or even worldwide. Success is marked by a thriving and inclusive user base.
- **Accessibility and Inclusivity:** Ensure that the QuizMasters Web and Mobile Application is accessible to individuals of varying abilities and backgrounds. Success is defined by providing equal opportunities for all participants.
- **Content Quality:** Maintain a high standard of quiz content in terms of accuracy, relevance, and variety. Success is characterized by consistently offering engaging and diverse topics.
- **Engagement Metrics:** Track engagement metrics such as participation rates, time spent on the platform, and interactions among participants. Success involves increasing user engagement over time.
- **Fair Play and Sportsmanship:** Ensure a fair and respectful competition environment by monitoring and addressing issues related to cheating, harassment, or unsportsmanlike behavior. Success is seen in a positive and respectful community.

- **Participant Satisfaction:** Continuously gather feedback from participants and assess their overall satisfaction with the platform. Success is reflected in high participant ratings and positive testimonials.

Future Updates

- **Knowledge Advancement:** Evaluate participants' knowledge improvement through quizzes and competitions. Success is evident when participants consistently display growth in their knowledge and problem-solving abilities.
- **Learning Impact:** Assess the educational impact of the QuizMasters Application on participants' academic and personal growth. Success is demonstrated through participants' academic achievements and personal development stories linked to the platform.
- **Global Reach:** Measure the expansion of the Quiz Bee community across geographical boundaries. Success is achieved when the platform gains popularity and active users from different countries and cultures.
- **Financial Viability:** Achieve financial sustainability by balancing revenue generation with affordability and accessibility. Success is maintaining the platform's financial health while keeping it accessible to a wide audience.

By adhering to this vision, mission, and success criteria, QuizMasters aims to create a vibrant and supportive community of learners and competitors, where participants can grow, connect, and celebrate the joy of learning together.

Website Development Process Checklist

1. Project Definition and Discovery

- Set website and mobile application goals
- Formulate a budget
- Do a competitive analysis
- Document project requirements
- Establish project team and roles

2. Content Strategy & Architecture

- Gather and assess content
- Create sitemap
- Write content
- Content review
- Content approval

3. Visual Design & Prototyping

- Create mood board
- Build wireframes
- Design page mockups
- Design review
- Design approval

4. Website and Mobile Application Build & Development

- Database - MS SQL
- For Web App
 - Backend: C#, ASP.Net Core, Microservices, API, RabbitMQ, gRPC, JWT
 - Frontend: Vite, React, Tailwind
- For Mobile App
 - React Native, NativeWind

5. Final Site Review & QA Testing

- Review full site
- QA site
- Make final updates
- Final approval

6. Website Launch & Promotion

- Official launch
- Site promotion

7. Maintenance & Monitoring

Goals

- **Engage and Entertain Users:** We want our platform to be a fun and engaging place for people to test their knowledge and enjoy quizzes on a wide range of topics.
- **Educate and Inform:** Our goal is to provide users with interesting and informative questions that help them learn new facts and expand their knowledge.
- **Content Variety:** We aim to offer a diverse range of quizzes, including trivia, educational, and themed quizzes to cater to different interests.
- **User-Friendly Experience:** We aim to create an easy-to-use website and mobile app that anyone can navigate effortlessly, regardless of their tech-savviness.
- **Gamification:** Implement gamification elements to make the quiz-taking experience more enjoyable and rewarding.
- **Accessibility:** It's important that our platform is accessible to people of all abilities, including those with disabilities.
- **Mobile Convenience:** We aim to provide a seamless mobile experience, allowing users to enjoy quizzes on the go.
- **Quality Assurance:** Our quizzes should be accurate, well-researched, and free from errors or biases.
- **Community Building:** We want to foster a sense of community among users, allowing them to compete, share their scores, and connect with others who share their interests.
- **Scalability:** Our platform should be capable of handling a growing number of users and quizzes without performance issues.
- **Transparency:** We will be transparent about how our platform operates, including how quiz topics are selected and how user data is used.
- **User Support:** Providing excellent customer support to address user inquiries and issues is a top priority.
- **Feedback and Improvement:** We are committed to actively listening to user feedback and continuously improving our quizzes and platform.
- **Continuous Innovation:** We will stay up-to-date with technology trends and continuously innovate to keep our platform fresh and engaging for users.

Future Goal Updates:

- **Personalized Content:** We plan to offer personalized quiz recommendations based on users' preferences and past quiz performance.
- **Social Integration:** Users should be able to easily share their quiz results and activities on social media platforms.
- **Monetization:** While our primary focus is on providing a great user experience, we also plan to generate revenue through advertising or premium subscription options.
- **Data Security and Privacy:** We will prioritize the security and privacy of user data, ensuring that their personal information is protected.
- **Global Reach:** We want to make our quizzes accessible to a global audience, with content that appeals to people from different cultural backgrounds.
- **Partnerships and Collaborations:** We may explore partnerships with content creators and organizations to expand our quiz offerings.

These goals are designed to ensure that QuizMaster becomes a beloved platform for quiz enthusiasts, fostering learning and fun in a welcoming and user-centric environment.

Features / Requirements

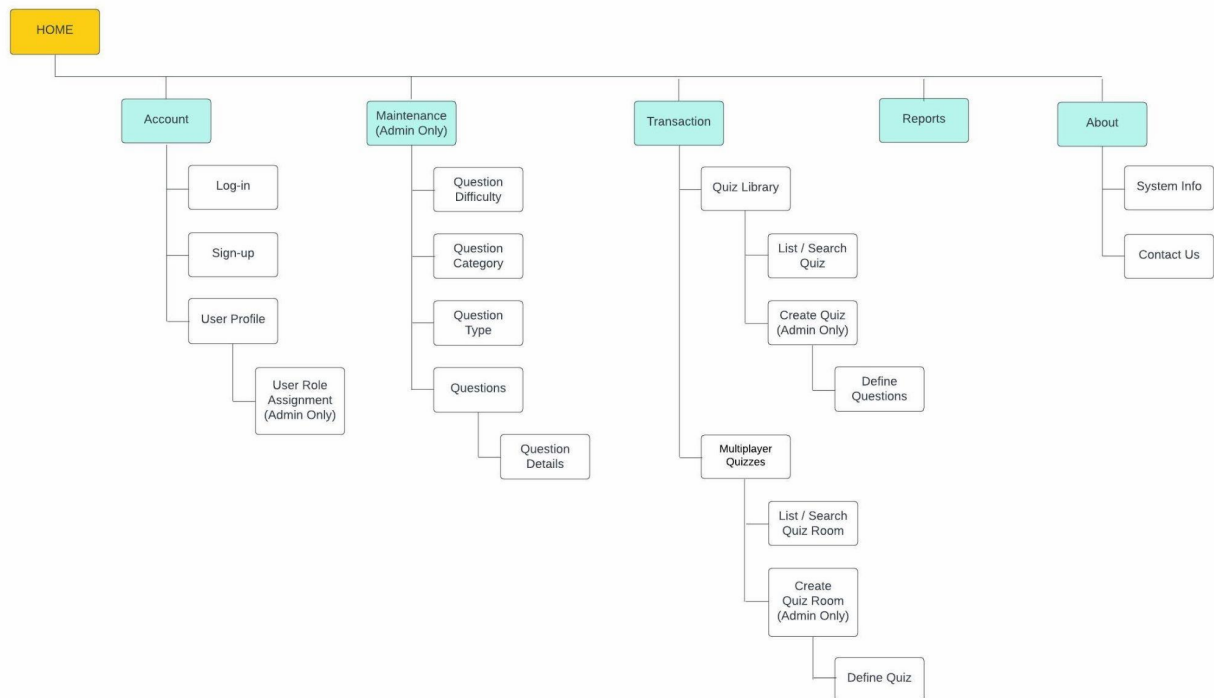
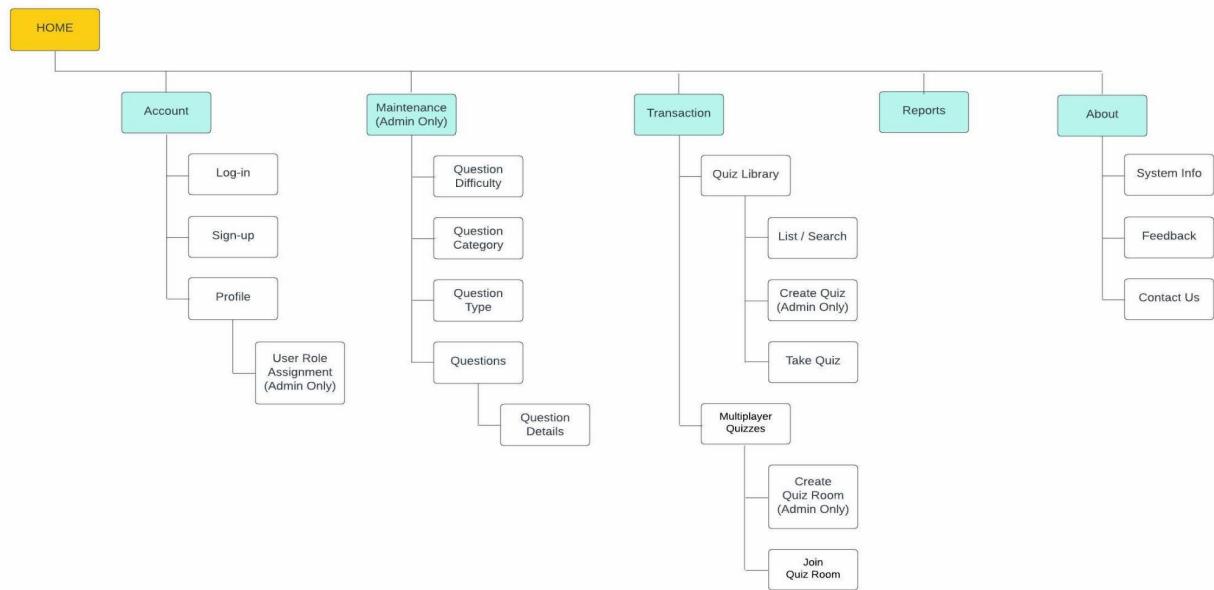
1. **User Roles and Permissions:** Define what users can and cannot do within the system, restricting access to sensitive data and features based on their assigned roles.
Administrator account type can access most of the modules or features of the system while the User account type is a limited access type of account.
2. **User Profiles:** Create your unique profile, showcase your quiz achievements, and connect with other quiz enthusiasts.
3. **Quiz Creation:** Craft your own quizzes and share them with the community to challenge their knowledge.
4. **Quiz Library:** Explore a vast collection of quizzes on topics you love, from movies and sports to history and science.
5. **Search and Filter:** Find the perfect quiz quickly with intuitive search and filtering options.
6. **Multiplayer Quizzes:** Challenge friends in real-time or join public multiplayer quizzes for interactive fun.
7. **Leaderboards:** Compete with friends and other users, climb the ranks, and see who's the ultimate QuizMaster.
8. **Gamified Experience:** Enjoy a game-like experience with achievements, badges, and rewards for your quiz success.
9. **Interactive Learning:** Discover informative quizzes that help you learn while having fun.
10. **Accessibility Features:** Our platform is designed to be inclusive, with features for users with different abilities.
11. **Customer Support:** Reach out to our friendly support team for any questions or assistance you may need.
12. **Feedback and Reporting:** Provide feedback on quizzes and report any issues or inaccuracies.
13. **In-App Messaging:** Stay connected with friends and fellow quizzers through in-app messaging.

Future Feature/Requirement Updates:

14. **Offline Mode:** Download quizzes to play offline when you don't have an internet connection.
15. **Commenting and Sharing:** Engage with others by leaving comments on quizzes and easily share your favorite quizzes on social media.
16. **Ad-Free Option:** Enjoy an ad-free experience with a premium subscription.
17. **Data Privacy Controls:** Easily manage your data and privacy settings to ensure your information is secure.
18. **Push Notifications:** Stay updated with notifications about new quizzes, challenges, and leaderboard changes.
19. **Personalized Recommendations:** Get quiz suggestions tailored to your interests and past quiz performance.
20. **Quiz Challenges:** Receive daily or weekly quiz challenges to keep your skills sharp.
21. **Quiz of the Day:** Start your day with a thought-provoking "Quiz of the Day."

These features are designed to make your experience on the QuizMaster platform enjoyable, interactive, and tailored to your preferences while fostering a sense of community among quiz enthusiasts.

Sitemap



Database Design

ASPNetCore.Identity (IdentityDbContext)

Note: Implementation of **AspNetCore.Identity** will result in the creation of the following relationship and entities.

AspNetRoleClaims

	Name	Data Type	Allow Nulls	Default
PK	Id	int	No	Identity
	RoleId	int	No	
	ClaimType	varchar(max)	Yes	
	ClaimValue	varchar(max)	Yes	

AspNetRoles

	Name	Data Type	Allow Nulls	Default
PK	Id	int	No	Identity
	UserRoleDesc	varchar(15)	No	
	Name	varchar(256)	Yes	
	NormalizedName	varchar(256)	Yes	
	ConcurrencyStamp	varchar(max)	Yes	

AspNetUserClaims

	Name	Data Type	Allow Nulls	Default
PK	Id	int	No	Identity
	UserId	int	No	
	ClaimType	varchar(max)	Yes	
	ClaimValue	varchar(max)	Yes	

AspNetUserLogins

	Name	Data Type	Allow Nulls	Default
PK	LoginProvider	varchar(450)	No	
PK	ProviderKey	varchar(450)	No	
FK	UserId	int	no	

	ProviderDisplayName	varchar(max)	Yes	
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AspNetUserRoles

	Name	Data Type	Allow Nulls	Default
FK	UserId	int	No	
FK	RoleId	int	No	

AspNetUsers

	Name	Data Type	Allow Nulls	Default
PK	UserId	int	No	Identity
	LastName	varchar(50)	Yes	
	FirstName	varchar(50)	Yes	
	Email	varchar(MAX)	No	
	UserName	varchar(20)	No	
	PasswordHash	varchar(MAX)	Yes	
FK	UserRoleId	int	No	
	ActiveData	bit	No	1
	DateCreated	datetime2	No	
	DateUpdated	datetime2	Yes	
	UpdatedByUserId	int	Yes	
	NormalizedUserName	varchar(MAX)	Yes	
	NormalizedEmail	varchar(MAX)	Yes	
	EmailConfirmed	bit	No	0
	SecurityStamp	varchar(MAX)	Yes	
	ConcurrencyStamp	varchar(MAX)	Yes	
	PhoneNumber	varchar(MAX)	Yes	
	PhoneNumberConfirmed	bit	No	0
	TwoFactorEnabled	bit	No	0

	LockoutEnd	datetimeoffset(7)	Yes	
	LockoutEnabled	bit	No	0
	AccessFailedCount	int	No	0

Notes:

For Partial User:

- It shall require an Email and Username
- Default value is hashed of string.Empty for the password of "On the go"/"partial" user

AspNetUserTokens

	Name	Data Type	Allow Nulls	Default
FK	UserId	int	No	
PK	LoginProvider	varchar(450)	No	
PK	Name	varchar(450)	No	
	Value	varchar(max)	Yes	

QuestionDifficulty

	Name	Data Type	Allow Nulls	Default
PK	Id	int	No	Identity
	QDifficultyDesc	varchar(10)	No	
	ActiveData	bit	No	1
	DateCreated	datetime2	No	
	DateUpdated	datetime2	Yes	
FK	CreatedByUserId	int	No	
FK	UpdatedByUserId	int	Yes	

This table could be pre-loaded with the following values:

1. Easy
2. Average

3. Difficult

QuestionCategory

	Name	Data Type	Allow Nulls	Default
PK	Id	int	No	Identity
	QCategoryDesc	varchar(50)	No	
	ActiveData	bit	No	1
	DateCreated	datetime2	No	
	DateUpdated	datetime2	Yes	
FK	CreatedByUserId	int	No	
FK	UpdatedByUserId	int	Yes	

Examples:

- Science
- Movies
- Animals
- Places
- People
- System Operations and Maintenance
- Data Structures
- Algorithms
- Etc.

QuestionType

	Name	Data Type	Allow Nulls	Default
PK	Id	int	No	Identity
	QTypeDesc	varchar(50)	No	
	QDetailRequired	bit	No	0
	ActiveData	bit	No	1
	DateCreated	datetime2	No	

	DateUpdated	datetime2	Yes	
	CreatedByUserId	int	No	
	UpdatedByUserId	int	Yes	

This table could be pre-loaded with the following values:

- *Multiple Choice (Give players several answer alternatives to choose from)
- *Multiple Choice + Audio (Write a quiz question to be read aloud)
- True or False (Let players decide if the statement is true or false)
- Type Answer (Ask players to type a short correct answer)
- *Slider (Let players guess the number on the slider)
- *Puzzle (Ask players to place answers in the correct order)

*Note: * Type with required question details*

Question

	Name	Data Type	Allow Nulls	Default
PK	QId	int	No	Identity
	QStatement	varchar(MAX)	No	
	QAnswer	varchar(MAX)	No	
	QImage	varchar(MAX)	No	
	QAudio	varchar(MAX)	Yes	
	QTime	int	No	
FK	QDifficultyId	int	No	
FK	QCategoryId	int	No	
FK	QTypeId	int	No	
	ActiveData	bit	No	1
	DateCreated	datetime2	No	
	DateUpdated	datetime2	Yes	
FK	CreatedByUserId	int	No	
FK	UpdatedByUserId	int	Yes	

QuestionDetail

	Name	Data Type	Allow Nulls	Default
PK	Id	int	No	Identity
	QDetailDesc	varchar(100)	No	
	QDetailValue	varChar(500)	No	
	QDetailDataType	varchar(50)	No	
FK	QuestionId	int	No	
	ActiveData	bit	No	1
	DateCreated	datetime2	No	
	DateUpdated	datetime2	Yes	
FK	CreatedByUserId	int	No	
FK	UpdatedByUserId	int	Yes	

*QDetailDesc: Option | TextToAudio | Minimum | Maximum | Interval

*QDetailValue: *Actual value of the QuestionDetail*

*QDetailDataType: "string"

QuestionAnswer

	Name	Data Type	Allow Nulls	Default
PK	Id	int	No	Identity
	QDetailDesc	varchar(100)	No	
	QDetailValue	varChar(500)	No	
	QDetailDataType	varchar(50)	No	
FK	QuestionId	int	No	
	ActiveData	bit	No	1
	DateCreated	datetime2	No	
	DateUpdated	datetime2	Yes	
FK	CreatedByUserId	int	No	

FK	UpdatedByUserId	int	Yes	

Set

	Name	Data Type	Allow Nulls	Default
PK	QSetId	int	No	Identity
	QSetName	varChar(50)	No	
	QSetDesc	varchar(MAX)	No	
	ActiveData	bit	No	1
	DateCreated	datetime2	No	
	DateUpdated	datetime2	Yes	
FK	CreatedByUserId	int	No	
FK	UpdatedByUserId	int	Yes	

QuestionSet

	Name	Data Type	Allow Nulls	Default
FK	QId	int	No	
FK	QSetId	int	No	
	ActiveData	bit	No	1
	DateCreated	datetime2	No	
	DateUpdated	datetime2	Yes	
FK	CreatedByUserId	int	No	
FK	UpdatedByUserId	int	Yes	

QuizRoom

	Name	Data Type	Allow Nulls	Default
PK	QRoomId	int	No	Identity
	QRoomDesc	varchar(MAX)	No	
	QRoomPin	int	No	
	RoomOptions	varchar(MAX)	Yes	
	ActiveData	bit	No	1
	DateCreated	datetime2	No	
	DateUpdated	datetime2	Yes	
FK	CreatedByUserId	int	No	
FK	UpdatedByUserId	int	Yes	

SetQuizRoom

	Name	Data Type	Allow Nulls	Default
FK	QSetId	int	No	
FK	QRoomId	int	No	
	ActiveData	bit	No	1
	DateCreated	datetime2	No	
	DateUpdated	datetime2	Yes	
FK	CreatedByUserId	int	No	
FK	UpdatedByUserId	int	Yes	

QuizParticipant

	Name	Data Type	Allow Nulls	Default
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PK	QParticipantId	int	No	Identity
	QParticipantDesc	varchar(50)	No	
FK	QRoomId	int	No	
FK	UserId	int	No	
	Score	int	No	
	QStartDate	datetime2	No	
	QEndDate	datetime2	Yes	
	QStatus	bit	No	0
	ActiveData	bit	No	1
	DateCreated	datetime2	No	
	DateUpdated	datetime2	Yes	
FK	CreatedByUserId	int	No	
FK	UpdatedByUserId	int	Yes	

QParticipantDesc - this should be a short description/name of the participant like school name

QuizDetail (Stored in Memory - Initial Design)

	Name	Data Type	Allow Nulls	Default
FK	QRoomId	int	No	
FK	QParticipantId	int	No	
FK	QId	int	No	
	Answer	varchar(MAX)	No	
	AnswerDetail	varchar(MAX)	No	
	ActiveData	bit	No	1
	DateCreated	datetime2	No	
	DateUpdated	datetime2	Yes	
FK	CreatedByUserId	int	No	

FK	UpdatedByUserId	int	Yes	

UserAuditTrail

	Name	Data Type	Allow Nulls	Default
FK	UserAuditTrailId	int	No	
	UserId	int	No	
	Action	varchar(MAX)	No	
	Timestamp	datetime2	No	
	Details	varchar(MAX)	No	
	UserRole	varchar(MAX)	No	
	OldValues	datetime2	Yes	
	NewValues	datetime2	Yes	

QuizAuditTrail

	Name	Data Type	Allow Nulls	Default
FK	QuizAuditTrailId	int	No	
	UserId	int	No	
	Action	varchar(MAX)	No	
	Timestamp	datetime2	No	
	Details	varchar(MAX)	No	
	UserRole	varchar(MAX)	No	
	OldValues	datetime2	Yes	
	NewValues	datetime2	Yes	

Implementation (Coding)

Git Commit Message Standards

Feature (Feat)

Use this category when you're adding a new feature to your software. A feature could be a new function, capability, or a significant enhancement to existing functionality.

```
Feat: Add authentication controller.
```

Bug Fix (Fix)

When you're fixing a bug or resolving an issue in the code, use this category. It's important to describe the issue briefly

```
Fix: Fix null pointer exception on authentication controller.
```

Refactor (Ref)

Use this category when you're making code improvements without changing its external behavior. This could involve code cleanup, optimization, or restructuring.

```
Ref: Make the filter function efficient.
```

Documentation (Doc)

When you're making changes to documentation, such as updating README files, inline code comments, or user guides.

```
Doc: Update installation instructions in README
```

Test (Test)

This category is for changes related to testing. It could involve adding new tests, modifying existing ones, or fixing issues related to testing.

```
Test: Add unit tests for the user registration module.
```

Git Branching Convention

main

Branch for production. Has branch protection rules. Requires Sir Rodney's approval .

develop (develop -> main)

Branch for development. Compilation of features and bug fixes. Has branch protection rules. Requires at least 2 approvals.

Feature (not a branch)- (feature -> develop -> main)

Branch for new or existing features, refactor, and enhancements that are currently in progress. Follow the branch naming convention below.

feature/{feature}

ex: feature/quizmaster-api-quiz

feature/quizmaster-frontend-web-maintenance

bugfix (not a branch)- (bugfix -> develop -> main)

Branch for large fixes. Follow the branch naming convention below.

bugfix/{bugfix}

ex: bugfix/fix-quizmaster-api-quiz

hotfix - (hotfix -> main)

Branch for quick fixes (e.g. few lines of code). Follow the branch naming convention below.

hotfix/{hotfix}

ex: hotfix/fix-text-color

Doc - (doc -> develop -> main)

Branch for project documentations

doc/{doc}

ex: doc/update-database

Test - (test -> develop -> main)

Branch for test.

test/{test}

ex: test/quizmaster-api-quiz